MATH 111 Calculus 1

Revision

Limits, continuity, the derivative — Lectures 1–10 in questions

Week 4 · Revision lecture · Stewart §1.5–1.8, §3.4, §2.1–2.6

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

While you wait

Sit with your team. Without notes, on one line each: the derivative of $x \sin x$; of $\sin 2x$; of $\sqrt{x}$. Next to each, the name of the rule you used. Then compare with your neighbour — and disagree if you can.

Two questions from Lecture 10

"Do I really need the chain rule for something as small as $\sin 2x$?"

Two questions from Lecture 10

"Do I really need the chain rule for something as small as $\sin 2x$?"

Not strictly — $\sin 2x = 2\sin x \cos x$ and the product rule get there too — but the chain rule does it in one line: outside sin, inside $2x$, derivative $\cos 2x \cdot 2 = 2\cos 2x$. That "2" is the inside's derivative, and leaving it out is one of the commonest slips in Chapter 2 — which is why it is on the title slide today.

Two questions from Lecture 10

"Do I really need the chain rule for something as small as $\sin 2x$?"

Not strictly — $\sin 2x = 2\sin x \cos x$ and the product rule get there too — but the chain rule does it in one line: outside sin, inside $2x$, derivative $\cos 2x \cdot 2 = 2\cos 2x$. That "2" is the inside's derivative, and leaving it out is one of the commonest slips in Chapter 2 — which is why it is on the title slide today.

"Why may I treat y as a function of x on the circle, when the circle is not a graph?"

Two questions from Lecture 10

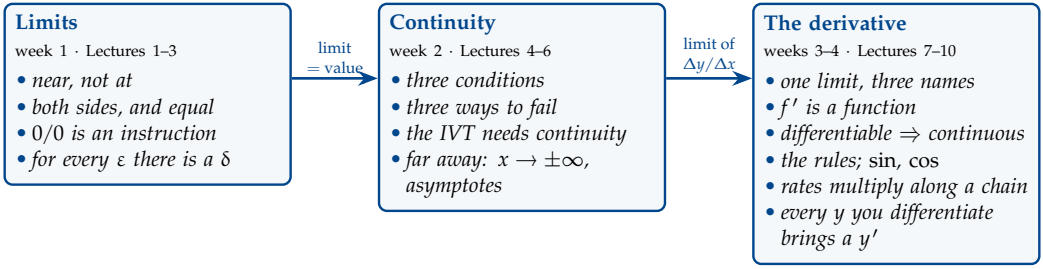
"Do I really need the chain rule for something as small as $\sin 2x$?"

Not strictly — $\sin 2x = 2\sin x \cos x$ and the product rule get there too — but the chain rule does it in one line: outside sin, inside $2x$, derivative $\cos 2x \cdot 2 = 2\cos 2x$. That "2" is the inside's derivative, and leaving it out is one of the commonest slips in Chapter 2 — which is why it is on the title slide today.

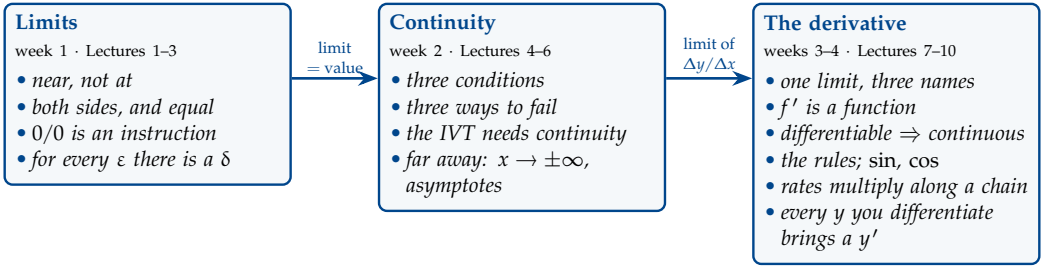
"Why may I treat y as a function of x on the circle, when the circle is not a graph?"

Because near the point it *is* one — the upper half or the lower half — and the method never asks which. Where it is not, at $(13, 0)$, the method says so itself: no y' solves the differentiated equation.

The map — the course so far on one page



The map — the course so far on one page



Today

Everything after week 1 is the limit, *used*: continuity is a limit equal to a value; the derivative, a limit of difference quotients; every rule, that limit computed once and for all.

STAND UP

"If $f(x) \to 5$ as $x \to 2$, then $f(2) = 5$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

STAND UP

"If $f(x) \to 5$ as $x \to 2$, then $f(2) = 5$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

Verdict

FALSE. The limit looks at x *near* 2, never at 2 itself: $f(2)$ may be 7, or undefined, and the limit is still 5. A limit and a value agree exactly when f is continuous at 2 — that is the definition of continuity, and it is the bridge from week 1 to week 2.

Question 23

$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - x - 6} = ?$$

A. 0 — the numerator is 0 at $x = 3$.

B. Does not exist — $\frac{0}{0}$.

C. $\frac{6}{5}$

D. 1

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - x - 6} = ?$$

A. 0 — the numerator is 0 at $x = 3$.

B. Does not exist — $\frac{0}{0}$.

C. $\frac{6}{5}$

D. 1



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

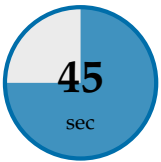
$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - x - 6} = ?$$

A. 0 — the numerator is 0 at $x = 3$.

B. Does not exist — $\frac{0}{0}$.

C. $\frac{6}{5}$

D. 1



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

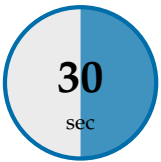
$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - x - 6} = ?$$

A. 0 — the numerator is 0 at $x = 3$.

B. Does not exist — $\frac{0}{0}$.

C. $\frac{6}{5}$

D. 1



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

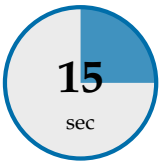
$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - x - 6} = ?$$

A. 0 — the numerator is 0 at $x = 3$.

B. Does not exist — $\frac{0}{0}$.

C. $\frac{6}{5}$

D. 1



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

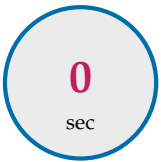
$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - x - 6} = ?$$

A. 0 — the numerator is 0 at $x = 3$.

B. Does not exist — $\frac{0}{0}$.

C. $\frac{6}{5}$

D. 1



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

VOTE

$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - x - 6} = ?$$

A. 0 — the numerator is 0 at $x = 3$.

B. Does not exist — $\frac{0}{0}$.

C. $\frac{6}{5}$

D. 1

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - x - 6} = ?$$

A. 0 — the numerator is 0 at $x = 3$.

B. Does not exist — $\frac{0}{0}$.

C. $\frac{6}{5}$

D. 1

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - x - 6} = ?$$

A. 0 — the numerator is 0 at $x = 3$.

B. Does not exist — $\frac{0}{0}$.

C. $\frac{6}{5}$

D. 1

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - x - 6} = ?$$

A. 0 — the numerator is 0 at $x = 3$.

B. Does not exist — $\frac{0}{0}$.

C. $\frac{6}{5}$

D. 1

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B **C** D

VOTE

$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - x - 6} = ?$$

A. 0 — the numerator is 0 at $x = 3$.

B. Does not exist — $\frac{0}{0}$.

C. $\frac{6}{5}$

D. 1

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C **D**

VOTE

$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - x - 6} = ?$$

A. 0 — the numerator is 0 at $x = 3$.

B. Does not exist — $\frac{0}{0}$.

C. $\frac{6}{5}$

D. 1

Hands down. Thank you.

A B C D

EXPLAIN

$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 - x - 6} = ?$$

A. 0 — the numerator is 0 at $x = 3$.

B. Does not exist — $\frac{0}{0}$.

C. $\frac{6}{5}$

D. 1

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

EXPLAIN

$$\lim_{x\to 3}\frac{x^2-9}{x^2-x-6}=?$$

A. 0 — the numerator is 0 at $x=3$.

B. Does not exist — $\frac{0}{0}$.

C. $\frac{6}{5}$

D. 1

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

C. $\frac{0}{0}$ is an instruction, not an answer: factor, $\frac{(x-3)(x+3)}{(x-3)(x+2)}=\frac{x+3}{x+2}$ for $x\neq 3$ — and the limit looks only at such x — so the limit is $\frac{6}{5}$. A reads the numerator and forgets the denominator is 0 too; B stops where the work starts; D is the reflex "$0/0=1$".

STAND UP

"In the definition of $\lim_{x \to a} f(x) = L$ you may choose δ first and then find an ε that works."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

STAND UP

"In the definition of $\lim_{x \to a} f(x) = L$ you may choose δ first and then find an ε that works."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

Verdict

FALSE. *For every ε there is a δ*: ε is the challenge, δ the response, and the order is the whole content of the definition. With the order reversed, every function bounded near a would pass, with *any* L: pick any δ, then choose ε larger than everything f does on that interval. Whoever moves second is the one who must answer.

Question 24

Let $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ for $x \neq 3$, and $f(3) = 4$. Which statement is true?

A. f is continuous at 3.

B. f is not continuous at 3; the discontinuity is removable.

C. f is not continuous at 3: the formula gives $\dfrac{0}{0}$ there.

D. f is not continuous at 3; the discontinuity is a jump.

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

Let $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ for $x \neq 3$, and $f(3) = 4$. Which statement is true?

A. f is continuous at 3.

B. f is not continuous at 3; the discontinuity is removable.

C. f is not continuous at 3: the formula gives $\dfrac{0}{0}$ there.

D. f is not continuous at 3; the discontinuity is a jump.



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

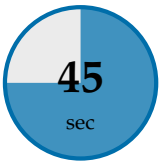
Let $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ for $x \neq 3$, and $f(3) = 4$. Which statement is true?

A. f is continuous at 3.

B. f is not continuous at 3; the discontinuity is removable.

C. f is not continuous at 3: the formula gives $\dfrac{0}{0}$ there.

D. f is not continuous at 3; the discontinuity is a jump.



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

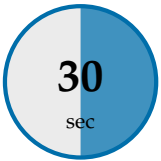
Let $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ for $x \neq 3$, and $f(3) = 4$. Which statement is true?

A. f is continuous at 3.

B. f is not continuous at 3; the discontinuity is removable.

C. f is not continuous at 3: the formula gives $\dfrac{0}{0}$ there.

D. f is not continuous at 3; the discontinuity is a jump.



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

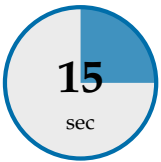
Let $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ for $x \neq 3$, and $f(3) = 4$. Which statement is true?

A. f is continuous at 3.

B. f is not continuous at 3; the discontinuity is removable.

C. f is not continuous at 3: the formula gives $\dfrac{0}{0}$ there.

D. f is not continuous at 3; the discontinuity is a jump.



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

Let $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ for $x \neq 3$, and $f(3) = 4$. Which statement is true?

A. f is continuous at 3.

B. f is not continuous at 3; the discontinuity is removable.

C. f is not continuous at 3: the formula gives $\dfrac{0}{0}$ there.

D. f is not continuous at 3; the discontinuity is a jump.



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

VOTE

Let $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ for $x \neq 3$, and $f(3) = 4$. Which statement is true?

A. f is continuous at 3.

B. f is not continuous at 3; the discontinuity is removable.

C. f is not continuous at 3: the formula gives $\dfrac{0}{0}$ there.

D. f is not continuous at 3; the discontinuity is a jump.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Let $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ for $x \neq 3$, and $f(3) = 4$. Which statement is true?

A. f is continuous at 3.

B. f is not continuous at 3; the discontinuity is removable.

C. f is not continuous at 3: the formula gives $\dfrac{0}{0}$ there.

D. f is not continuous at 3; the discontinuity is a jump.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Let $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ for $x \neq 3$, and $f(3) = 4$. Which statement is true?

A. f is continuous at 3.

B. f is not continuous at 3; the discontinuity is removable.

C. f is not continuous at 3: the formula gives $\dfrac{0}{0}$ there.

D. f is not continuous at 3; the discontinuity is a jump.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Let $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ for $x \neq 3$, and $f(3) = 4$. Which statement is true?

A. f is continuous at 3.

B. f is not continuous at 3; the discontinuity is removable.

C. f is not continuous at 3: the formula gives $\dfrac{0}{0}$ there.

D. f is not continuous at 3; the discontinuity is a jump.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B **C** D

VOTE

Let $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ for $x \neq 3$, and $f(3) = 4$. Which statement is true?

A. f is continuous at 3.

B. f is not continuous at 3; the discontinuity is removable.

C. f is not continuous at 3: the formula gives $\dfrac{0}{0}$ there.

D. f is not continuous at 3; the discontinuity is a jump.

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Let $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ for $x \neq 3$, and $f(3) = 4$. Which statement is true?

A. f is continuous at 3.

B. f is not continuous at 3; the discontinuity is removable.

C. f is not continuous at 3: the formula gives $\dfrac{0}{0}$ there.

D. f is not continuous at 3; the discontinuity is a jump.

Hands down. Thank you.

A B C D

Question 24

EXPLAIN

Let $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ for $x \neq 3$, and $f(3) = 4$. Which statement is true?

A. f is continuous at 3.

B. f is not continuous at 3; the discontinuity is removable.

C. f is not continuous at 3: the formula gives $\dfrac{0}{0}$ there.

D. f is not continuous at 3; the discontinuity is a jump.

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Question 24

EXPLAIN

Let $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ for $x \neq 3$, and $f(3) = 4$. Which statement is true?

A. f is continuous at 3.

B. f is not continuous at 3; the discontinuity is removable.

C. f is not continuous at 3: the formula gives $\dfrac{0}{0}$ there.

D. f is not continuous at 3; the discontinuity is a jump.

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

A. Three conditions: $f(3)$ exists ($= 4$); the limit exists — for $x \neq 3$, $f(x) = \dfrac{(x-3)(x+1)}{x-3} = x + 1 \to 4$; and they are equal. Continuous. C is the trap: $\dfrac{0}{0}$ at 3 is what the *formula* says, and the formula is not used at 3 — $0/0$ is an instruction, and here the instruction leads to 4. B would need the limit and the value to differ; D two different one-sided limits.

STAND UP

"f is continuous on $[0, 4]$, $f(0) = 3$ and $f(4) = 7$. So f has no zero on $(0, 4)$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

STAND UP

"f is continuous on $[0,4]$, $f(0)=3$ and $f(4)=7$. So f has no zero on $(0,4)$."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

Verdict

FALSE. The Intermediate Value Theorem says what *must* happen — every value between 3 and 7 is taken somewhere in $(0,4)$ — and nothing about what cannot: f may dip to -1 in the middle and come back. Draw it. A theorem's conclusion is a guarantee, not a fence.

Question 25

$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2 + 4}} = ?$$

A. 2

B. Does not exist.

C. 0

D. -2

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2+4}} = ?$$

A. 2

B. Does not exist.

C. 0

D. -2

60 sec

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

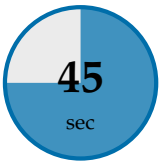
$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2+4}} = ?$$

A. 2

B. Does not exist.

C. 0

D. -2



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

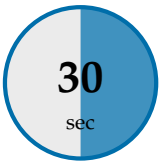
$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2+4}} = ?$$

A. 2

B. Does not exist.

C. 0

D. -2



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

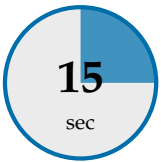
$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2 + 4}} = ?$$

A. 2

B. Does not exist.

C. 0

D. -2



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2+4}} = ?$$

A. 2

B. Does not exist.

C. 0

D. -2



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

VOTE

$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2+4}} = ?$$

A. 2

B. Does not exist.

C. 0

D. -2

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2+4}} = ?$$

A. 2

B. Does not exist.

C. 0

D. -2

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2+4}} = ?$$

A. 2

B. Does not exist.

C. 0

D. -2

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2+4}} = ?$$

A. 2

B. Does not exist.

C. 0

D. -2

Eyes on the screen. Hands up when your letter appears — hold for three seconds.



VOTE

$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2+4}} = ?$$

A. 2

B. Does not exist.

C. 0

D. -2

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C **D**

VOTE

$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2+4}} = ?$$

A. 2

B. Does not exist.

C. 0

D. -2

Hands down. Thank you.

A B C D

EXPLAIN

$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2 + 4}} = ?$$

A. 2

B. Does not exist.

C. 0

D. -2

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

EXPLAIN

$$\lim_{x\to-\infty} \frac{2x}{\sqrt{x^2+4}} = ?$$

A. 2

B. Does not exist.

C. 0

D. -2

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

D. Divide top and bottom by the highest power — but for $x < 0$, $\sqrt{x^2} = |x| = -x$, so $\sqrt{x^2+4} = -x\sqrt{1+4/x^2}$ and the quotient is $\dfrac{2}{-\sqrt{1+4/x^2}} \to -2$. A is the sign trap of Lecture 6: the answer at $+\infty$. The graph has two horizontal asymptotes, $y = 2$ and $y = -2$, one at each end.

Question 26

Prove from the definition that $\lim_{x \to 1} (10 - 7x) = 3$. The proof begins: "*Let* $\varepsilon > 0$. *Choose* $\delta = \ldots$" Which δ makes the proof work?

A. $\delta = 7\varepsilon$

B. $\delta = \dfrac{\varepsilon}{7}$

C. $\delta = \varepsilon$

D. $\delta = -\dfrac{\varepsilon}{7}$

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

Prove from the definition that $\lim_{x \to 1} (10 - 7x) = 3$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ makes the proof work?

A. $\delta = 7\varepsilon$

B. $\delta = \dfrac{\varepsilon}{7}$

C. $\delta = \varepsilon$

D. $\delta = -\dfrac{\varepsilon}{7}$



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

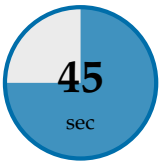
Prove from the definition that $\lim_{x \to 1} (10 - 7x) = 3$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ makes the proof work?

A. $\delta = 7\varepsilon$

B. $\delta = \dfrac{\varepsilon}{7}$

C. $\delta = \varepsilon$

D. $\delta = -\dfrac{\varepsilon}{7}$



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

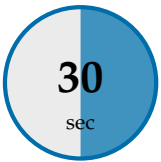
Prove from the definition that $\lim_{x \to 1} (10 - 7x) = 3$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ makes the proof work?

A. $\delta = 7\varepsilon$

B. $\delta = \dfrac{\varepsilon}{7}$

C. $\delta = \varepsilon$

D. $\delta = -\dfrac{\varepsilon}{7}$



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

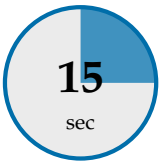
Prove from the definition that $\lim_{x \to 1} (10 - 7x) = 3$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ makes the proof work?

A. $\delta = 7\varepsilon$

B. $\delta = \dfrac{\varepsilon}{7}$

C. $\delta = \varepsilon$

D. $\delta = -\dfrac{\varepsilon}{7}$



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

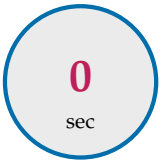
Prove from the definition that $\lim_{x \to 1} (10 - 7x) = 3$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ makes the proof work?

A. $\delta = 7\varepsilon$

B. $\delta = \dfrac{\varepsilon}{7}$

C. $\delta = \varepsilon$

D. $\delta = -\dfrac{\varepsilon}{7}$



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

VOTE

Prove from the definition that $\lim_{x \to 1} (10 - 7x) = 3$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ makes the proof work?

A. $\delta = 7\varepsilon$

B. $\delta = \dfrac{\varepsilon}{7}$

C. $\delta = \varepsilon$

D. $\delta = -\dfrac{\varepsilon}{7}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Prove from the definition that $\lim_{x \to 1} (10 - 7x) = 3$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ makes the proof work?

A. $\delta = 7\varepsilon$

B. $\delta = \dfrac{\varepsilon}{7}$

C. $\delta = \varepsilon$

D. $\delta = -\dfrac{\varepsilon}{7}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Prove from the definition that $\lim_{x \to 1} (10 - 7x) = 3$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ makes the proof work?

A. $\delta = 7\varepsilon$

B. $\delta = \dfrac{\varepsilon}{7}$

C. $\delta = \varepsilon$

D. $\delta = -\dfrac{\varepsilon}{7}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Prove from the definition that $\lim_{x \to 1} (10 - 7x) = 3$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ makes the proof work?

A. $\delta = 7\varepsilon$

B. $\delta = \dfrac{\varepsilon}{7}$

C. $\delta = \varepsilon$

D. $\delta = -\dfrac{\varepsilon}{7}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B **C** D

VOTE

Prove from the definition that $\lim_{x \to 1} (10 - 7x) = 3$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ makes the proof work?

A. $\delta = 7\varepsilon$

B. $\delta = \dfrac{\varepsilon}{7}$

C. $\delta = \varepsilon$

D. $\delta = -\dfrac{\varepsilon}{7}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C **D**

VOTE

Prove from the definition that $\lim_{x \to 1} (10 - 7x) = 3$. The proof begins: *"Let* $\varepsilon > 0$. *Choose* $\delta = \ldots$*"* Which δ makes the proof work?

A. $\delta = 7\varepsilon$

B. $\delta = \dfrac{\varepsilon}{7}$

C. $\delta = \varepsilon$

D. $\delta = -\dfrac{\varepsilon}{7}$

Hands down. Thank you.

A B C D

EXPLAIN

Prove from the definition that $\lim_{x \to 1} (10 - 7x) = 3$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ makes the proof work?

A. $\delta = 7\varepsilon$

B. $\delta = \dfrac{\varepsilon}{7}$

C. $\delta = \varepsilon$

D. $\delta = -\dfrac{\varepsilon}{7}$

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

EXPLAIN

Prove from the definition that $\lim_{x \to 1} (10 - 7x) = 3$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ makes the proof work?

A. $\delta = 7\varepsilon$

B. $\delta = \dfrac{\varepsilon}{7}$

C. $\delta = \varepsilon$

D. $\delta = -\dfrac{\varepsilon}{7}$

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

B. $|(10 - 7x) - 3| = |7 - 7x| = 7\,|x - 1|$, and that is $< \varepsilon$ exactly when $|x - 1| < \varepsilon/7$. A turns the scaling upside down: $|x - 1| < 7\varepsilon$ lets $f(x)$ move up to 49ε away from 3. C forgets the 7: $f(x)$ moves seven times as far as x. D is not allowed: the definition asks for $\delta > 0$ — the minus sign of $-7x$ is gone inside the absolute value.

Question 26 — the proof, written out

Scratch work — backwards, from what we want

We want $|f(x) - 3| < \varepsilon$. $\quad |(10 - 7x) - 3| = |7 - 7x| = 7\,|x - 1|$, and
$7\,|x - 1| < \varepsilon \iff |x - 1| < \dfrac{\varepsilon}{7}$. $\quad$ So: $\delta = \dfrac{\varepsilon}{7}$.

Question 26 — the proof, written out

Scratch work — backwards, from what we want

We want $|f(x) - 3| < \varepsilon$. $|(10 - 7x) - 3| = |7 - 7x| = 7\,|x - 1|$, and $7\,|x - 1| < \varepsilon \iff |x - 1| < \dfrac{\varepsilon}{7}$. So: $\delta = \dfrac{\varepsilon}{7}$.

Proof. Let $\varepsilon > 0$. Choose $\delta = \dfrac{\varepsilon}{7}$; then $\delta > 0$.

Question 26 — the proof, written out

Scratch work — backwards, from what we want

We want $|f(x) - 3| < \varepsilon$. $\quad |(10 - 7x) - 3| = |7 - 7x| = 7\,|x - 1|$, and
$7\,|x - 1| < \varepsilon \iff |x - 1| < \dfrac{\varepsilon}{7}$. $\quad$ So: $\delta = \dfrac{\varepsilon}{7}$.

Proof. Let $\varepsilon > 0$. Choose $\delta = \dfrac{\varepsilon}{7}$; then $\delta > 0$.

If $0 < |x - 1| < \delta$, then

$$|(10 - 7x) - 3| \;=\; 7\,|x - 1| \;<\; 7\delta \;=\; \varepsilon.$$

Question 26 — the proof, written out

Scratch work — backwards, from what we want

We want $|f(x) - 3| < \varepsilon$. $|(10 - 7x) - 3| = |7 - 7x| = 7\,|x - 1|$, and $7\,|x - 1| < \varepsilon \iff |x - 1| < \dfrac{\varepsilon}{7}$. So: $\delta = \dfrac{\varepsilon}{7}$.

Proof. Let $\varepsilon > 0$. Choose $\delta = \dfrac{\varepsilon}{7}$; then $\delta > 0$.

If $0 < |x - 1| < \delta$, then

$$|(10 - 7x) - 3| \;=\; 7\,|x - 1| \;<\; 7\delta \;=\; \varepsilon.$$

Hence $\lim\limits_{x \to 1} (10 - 7x) = 3$. □

Question 26 — the proof, written out

Scratch work — backwards, from what we want

We want $|f(x) - 3| < \varepsilon$. $\quad |(10 - 7x) - 3| = |7 - 7x| = 7\,|x - 1|$, and
$7\,|x - 1| < \varepsilon \iff |x - 1| < \dfrac{\varepsilon}{7}$. $\quad$ So: $\delta = \dfrac{\varepsilon}{7}$.

Proof. Let $\varepsilon > 0$. Choose $\delta = \dfrac{\varepsilon}{7}$; then $\delta > 0$.

If $0 < |x - 1| < \delta$, then

$$|(10 - 7x) - 3| \;=\; 7\,|x - 1| \;<\; 7\delta \;=\; \varepsilon.$$

Hence $\lim\limits_{x \to 1} (10 - 7x) = 3$. □

Four things make it a proof: "let $\varepsilon > 0$" first; a δ that depends on ε; the chain, ending in exactly ε; the concluding sentence. The scratch work finds δ — write it too, but the proof is the forward part. A smaller δ works too: smaller is always safe.

Question 27

Prove that $f(x) = x^2 + x$ is continuous at 2 — from the definition: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - 2| < \delta \Rightarrow |f(x) - f(2)| < \varepsilon$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{6}$

B. $\delta = \dfrac{\varepsilon}{|x+3|}$

C. $\delta = \min\left\{1, \dfrac{\varepsilon}{6}\right\}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\}$

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

Prove that $f(x) = x^2 + x$ is continuous at 2 — from the definition: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - 2| < \delta \Rightarrow |f(x) - f(2)| < \varepsilon$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{6}$

B. $\delta = \dfrac{\varepsilon}{|x+3|}$

C. $\delta = \min\left\{1, \dfrac{\varepsilon}{6}\right\}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\}$



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

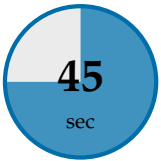
Prove that $f(x) = x^2 + x$ is continuous at 2 — from the definition: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - 2| < \delta \Rightarrow |f(x) - f(2)| < \varepsilon$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{6}$

B. $\delta = \dfrac{\varepsilon}{|x+3|}$

C. $\delta = \min\left\{1, \dfrac{\varepsilon}{6}\right\}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\}$



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

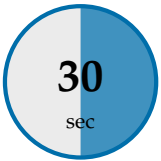
Prove that $f(x) = x^2 + x$ is continuous at 2 — from the definition: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - 2| < \delta \Rightarrow |f(x) - f(2)| < \varepsilon$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{6}$

B. $\delta = \dfrac{\varepsilon}{|x+3|}$

C. $\delta = \min\left\{1, \dfrac{\varepsilon}{6}\right\}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\}$



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

Prove that $f(x) = x^2 + x$ is continuous at 2 — from the definition: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - 2| < \delta \Rightarrow |f(x) - f(2)| < \varepsilon$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{6}$

B. $\delta = \dfrac{\varepsilon}{|x+3|}$

C. $\delta = \min\left\{1, \dfrac{\varepsilon}{6}\right\}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\}$



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

Prove that $f(x) = x^2 + x$ is continuous at 2 — from the definition: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - 2| < \delta \Rightarrow |f(x) - f(2)| < \varepsilon$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{6}$

B. $\delta = \dfrac{\varepsilon}{|x+3|}$

C. $\delta = \min\left\{1, \dfrac{\varepsilon}{6}\right\}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\}$

0
sec

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

VOTE

Prove that $f(x) = x^2 + x$ is continuous at 2 — from the definition: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - 2| < \delta \Rightarrow |f(x) - f(2)| < \varepsilon$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{6}$

B. $\delta = \dfrac{\varepsilon}{|x+3|}$

C. $\delta = \min\left\{1, \dfrac{\varepsilon}{6}\right\}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Prove that $f(x) = x^2 + x$ is continuous at 2 — from the definition: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - 2| < \delta \Rightarrow |f(x) - f(2)| < \varepsilon$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{6}$

B. $\delta = \dfrac{\varepsilon}{|x+3|}$

C. $\delta = \min\left\{1, \dfrac{\varepsilon}{6}\right\}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Prove that $f(x) = x^2 + x$ is continuous at 2 — from the definition: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - 2| < \delta \Rightarrow |f(x) - f(2)| < \varepsilon$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{6}$

B. $\delta = \dfrac{\varepsilon}{|x+3|}$

C. $\delta = \min\left\{1, \dfrac{\varepsilon}{6}\right\}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Prove that $f(x) = x^2 + x$ is continuous at 2 — from the definition: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - 2| < \delta \Rightarrow |f(x) - f(2)| < \varepsilon$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{6}$

B. $\delta = \dfrac{\varepsilon}{|x+3|}$

C. $\delta = \min\left\{1, \dfrac{\varepsilon}{6}\right\}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B **C** D

VOTE

Prove that $f(x) = x^2 + x$ is continuous at 2 — from the definition: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - 2| < \delta \Rightarrow |f(x) - f(2)| < \varepsilon$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{6}$

B. $\delta = \dfrac{\varepsilon}{|x+3|}$

C. $\delta = \min\left\{1, \dfrac{\varepsilon}{6}\right\}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C **D**

VOTE

Prove that $f(x) = x^2 + x$ is continuous at 2 — from the definition: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - 2| < \delta \Rightarrow |f(x) - f(2)| < \varepsilon$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{6}$

B. $\delta = \dfrac{\varepsilon}{|x+3|}$

C. $\delta = \min\left\{1, \dfrac{\varepsilon}{6}\right\}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\}$

Hands down. Thank you.

A B C D

EXPLAIN

Prove that $f(x) = x^2 + x$ is continuous at 2 — from the definition: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x-2| < \delta \Rightarrow |f(x) - f(2)| < \varepsilon$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{6}$

B. $\delta = \dfrac{\varepsilon}{|x+3|}$

C. $\delta = \min\left\{1, \dfrac{\varepsilon}{6}\right\}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\}$

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

EXPLAIN

Prove that $f(x) = x^2 + x$ is continuous at 2 — from the definition: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x-2| < \delta \Rightarrow |f(x) - f(2)| < \varepsilon$. The proof begins: *"Let $\varepsilon > 0$. Choose $\delta = \ldots$"* Which δ works for *every* $\varepsilon > 0$?

A. $\delta = \dfrac{\varepsilon}{6}$

B. $\delta = \dfrac{\varepsilon}{|x+3|}$

C. $\delta = \min\left\{1, \dfrac{\varepsilon}{6}\right\}$

D. $\delta = \min\left\{1, \dfrac{\varepsilon}{5}\right\}$

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

C. $|f(x) - f(2)| = |x^2 + x - 6| = |x-2|\,|x+3|$, and the second factor is not a constant: $\delta \le 1$ keeps x between 1 and 3, where $|x+3| < 6$; $\delta \le \varepsilon/6$ does the rest. A forgets the 1: for $\varepsilon = 60$ it lets in $x = 11$, where $|f(x) - 6| = 126$. B lets δ depend on x — but δ is chosen before any x. D freezes $|x+3|$ at its value 5 at $x = 2$: for $\varepsilon = 5$ it lets in $x = 2.9$, where $|f(x) - 6| = 5.31$.

Question 27 — the proof, written out

What must be shown (Lecture 5): for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x - 2| < \delta \;\Rightarrow\; |f(x) - f(2)| < \varepsilon$. Here $f(2) = 6$, and $x = 2$ itself is let in.

Question 27 — the proof, written out

What must be shown (Lecture 5): for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x-2| < \delta \;\Rightarrow\; |f(x)-f(2)| < \varepsilon$. Here $f(2) = 6$, and $x = 2$ itself is let in.

Scratch work

$|f(x)-6| = |x^2+x-6| = |x-2|\,|x+3|$. Tame the factor that is not a constant: if $|x-2| < 1$, then $1 < x < 3$, so $4 < x+3 < 6$ and $|x+3| < 6$; then $|f(x)-6| \le 6\,|x-2| < \varepsilon$ as soon as $|x-2| < \frac{\varepsilon}{6}$. Both conditions: $\delta = \min\{1, \frac{\varepsilon}{6}\}$.

Question 27 — the proof, written out

What must be shown (Lecture 5): for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x-2| < \delta \;\Rightarrow\; |f(x)-f(2)| < \varepsilon$. Here $f(2) = 6$, and $x = 2$ itself is let in.

Scratch work

$|f(x)-6| = |x^2+x-6| = |x-2|\,|x+3|$. Tame the factor that is not a constant: if $|x-2| < 1$, then $1 < x < 3$, so $4 < x+3 < 6$ and $|x+3| < 6$; then $|f(x)-6| \le 6\,|x-2| < \varepsilon$ as soon as $|x-2| < \frac{\varepsilon}{6}$. Both conditions: $\delta = \min\{1, \frac{\varepsilon}{6}\}$.

Proof. Let $\varepsilon > 0$. Choose $\delta = \min\{1, \frac{\varepsilon}{6}\}$; then $\delta > 0$.

Question 27 — the proof, written out

What must be shown (Lecture 5): for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x-2| < \delta \;\Rightarrow\; |f(x)-f(2)| < \varepsilon$. Here $f(2) = 6$, and $x = 2$ itself is let in.

Scratch work

$|f(x)-6| = |x^2+x-6| = |x-2|\,|x+3|$. Tame the factor that is not a constant: if $|x-2| < 1$, then $1 < x < 3$, so $4 < x+3 < 6$ and $|x+3| < 6$; then $|f(x)-6| \le 6\,|x-2| < \varepsilon$ as soon as $|x-2| < \frac{\varepsilon}{6}$. Both conditions: $\delta = \min\{1, \frac{\varepsilon}{6}\}$.

Proof. Let $\varepsilon > 0$. Choose $\delta = \min\{1, \frac{\varepsilon}{6}\}$; then $\delta > 0$.

If $|x-2| < \delta$, then $|x-2| < 1$, so $|x+3| < 6$; and $|x-2| < \frac{\varepsilon}{6}$. Hence

$$|f(x)-f(2)| \;=\; |x-2|\,|x+3| \;\le\; 6\,|x-2| \;<\; 6 \cdot \frac{\varepsilon}{6} \;=\; \varepsilon.$$

Question 27 — the proof, written out

What must be shown (Lecture 5): for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|x-2| < \delta \;\Rightarrow\; |f(x)-f(2)| < \varepsilon$. Here $f(2) = 6$, and $x = 2$ itself is let in.

Scratch work

$|f(x)-6| = |x^2+x-6| = |x-2|\,|x+3|$. Tame the factor that is not a constant: if $|x-2| < 1$, then $1 < x < 3$, so $4 < x+3 < 6$ and $|x+3| < 6$; then $|f(x)-6| \le 6\,|x-2| < \varepsilon$ as soon as $|x-2| < \frac{\varepsilon}{6}$. Both conditions: $\delta = \min\{1, \frac{\varepsilon}{6}\}$.

Proof. Let $\varepsilon > 0$. Choose $\delta = \min\{1, \frac{\varepsilon}{6}\}$; then $\delta > 0$.

If $|x-2| < \delta$, then $|x-2| < 1$, so $|x+3| < 6$; and $|x-2| < \frac{\varepsilon}{6}$. Hence

$$|f(x)-f(2)| \;=\; |x-2|\,|x+3| \;\le\; 6\,|x-2| \;<\; 6 \cdot \frac{\varepsilon}{6} \;=\; \varepsilon.$$

So f is continuous at 2. □

STAND UP

"If f is differentiable at a, then f is continuous at a."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

"If f is differentiable at a, then f is continuous at a."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

Verdict

TRUE — Lecture 8's theorem, with its two-line proof: for $x \neq a$, $f(x) - f(a) = \dfrac{f(x)-f(a)}{x-a} \cdot (x-a) \to f'(a) \cdot 0 = 0$. The *converse* is FALSE: $|x|$ at 0 is continuous, and its one-sided slopes are -1 and 1. Differentiable implies continuous, never the other way round.

Question 28

Error hunt. Compute $\dfrac{d}{dx}\sin^2(3x)$. Which line is the *first* wrong one?

Line 1: $\sin^2(3x) = (\sin 3x)^2$: outside the square, inside $\sin 3x$; so $2\sin 3x \cdot (\sin 3x)'$.

Line 2: $(\sin 3x)' = \cos 3x$.

Line 3: So the derivative is $2\sin 3x \cos 3x$.

Line 4: $= \sin 6x$ (double angle).

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

Error hunt. Compute $\dfrac{d}{dx}\sin^2(3x)$. Which line is the *first* wrong one?

Line 1: $\sin^2(3x) = (\sin 3x)^2$: outside the square, inside $\sin 3x$; so $2\sin 3x \cdot (\sin 3x)'$.

Line 2: $(\sin 3x)' = \cos 3x$.

Line 3: So the derivative is $2\sin 3x \cos 3x$.

Line 4: $= \sin 6x$ (double angle).

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

Error hunt. Compute $\dfrac{d}{dx}\sin^2(3x)$. Which line is the *first* wrong one?

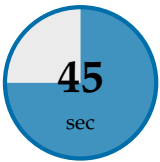
Line 1: $\sin^2(3x) = (\sin 3x)^2$: outside the square, inside $\sin 3x$; so $2\sin 3x \cdot (\sin 3x)'$.

Line 2: $(\sin 3x)' = \cos 3x$.

Line 3: So the derivative is $2\sin 3x \cos 3x$.

Line 4: $= \sin 6x$ (double angle).

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

Error hunt. Compute $\dfrac{d}{dx}\sin^2(3x)$. Which line is the *first* wrong one?

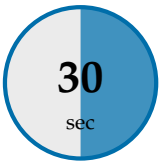
Line 1: $\sin^2(3x) = (\sin 3x)^2$: outside the square, inside $\sin 3x$; so $2\sin 3x \cdot (\sin 3x)'$.

Line 2: $(\sin 3x)' = \cos 3x$.

Line 3: So the derivative is $2\sin 3x \cos 3x$.

Line 4: $= \sin 6x$ (double angle).

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

Error hunt. Compute $\dfrac{d}{dx}\sin^2(3x)$. Which line is the *first* wrong one?

Line 1: $\sin^2(3x) = (\sin 3x)^2$: outside the square, inside $\sin 3x$; so $2\sin 3x \cdot (\sin 3x)'$.

Line 2: $(\sin 3x)' = \cos 3x$.

Line 3: So the derivative is $2\sin 3x \cos 3x$.

Line 4: $= \sin 6x$ (double angle).

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

Error hunt. Compute $\dfrac{d}{dx}\sin^2(3x)$. Which line is the *first* wrong one?

Line 1: $\sin^2(3x) = (\sin 3x)^2$: outside the square, inside $\sin 3x$; so $2\sin 3x \cdot (\sin 3x)'$.

Line 2: $(\sin 3x)' = \cos 3x$.

Line 3: So the derivative is $2\sin 3x \cos 3x$.

Line 4: $= \sin 6x$ (double angle).

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

VOTE

Error hunt. Compute $\dfrac{d}{dx}\sin^2(3x)$. Which line is the *first* wrong one?

Line 1: $\sin^2(3x) = (\sin 3x)^2$: outside the square, inside $\sin 3x$; so $2\sin 3x \cdot (\sin 3x)'$.
Line 2: $(\sin 3x)' = \cos 3x$.
Line 3: So the derivative is $2\sin 3x \cos 3x$.
Line 4: $= \sin 6x$ (double angle).

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Error hunt. Compute $\dfrac{d}{dx}\sin^2(3x)$. Which line is the *first* wrong one?

Line 1: $\sin^2(3x) = (\sin 3x)^2$: outside the square, inside $\sin 3x$; so $2\sin 3x \cdot (\sin 3x)'$.
Line 2: $(\sin 3x)' = \cos 3x$.
Line 3: So the derivative is $2\sin 3x \cos 3x$.
Line 4: $= \sin 6x$ (double angle).

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Error hunt. Compute $\dfrac{d}{dx}\sin^2(3x)$. Which line is the *first* wrong one?

Line 1: $\sin^2(3x) = (\sin 3x)^2$: outside the square, inside $\sin 3x$; so $2\sin 3x \cdot (\sin 3x)'$.
Line 2: $(\sin 3x)' = \cos 3x$.
Line 3: So the derivative is $2\sin 3x \cos 3x$.
Line 4: $= \sin 6x$ (double angle).

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Error hunt. Compute $\dfrac{d}{dx}\sin^2(3x)$. Which line is the *first* wrong one?

Line 1: $\sin^2(3x) = (\sin 3x)^2$: outside the square, inside $\sin 3x$; so $2\sin 3x \cdot (\sin 3x)'$.
Line 2: $(\sin 3x)' = \cos 3x$.
Line 3: So the derivative is $2\sin 3x \cos 3x$.
Line 4: $= \sin 6x$ (double angle).

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B **C** D

VOTE

Error hunt. Compute $\dfrac{d}{dx}\sin^2(3x)$. Which line is the *first* wrong one?

Line 1: $\sin^2(3x) = (\sin 3x)^2$: outside the square, inside $\sin 3x$; so $2\sin 3x \cdot (\sin 3x)'$.
Line 2: $(\sin 3x)' = \cos 3x$.
Line 3: So the derivative is $2\sin 3x \cos 3x$.
Line 4: $= \sin 6x$ (double angle).

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

Error hunt. Compute $\dfrac{d}{dx}\sin^2(3x)$. Which line is the *first* wrong one?

Line 1: $\sin^2(3x) = (\sin 3x)^2$: outside the square, inside $\sin 3x$; so $2\sin 3x \cdot (\sin 3x)'$.
Line 2: $(\sin 3x)' = \cos 3x$.
Line 3: So the derivative is $2\sin 3x \cos 3x$.
Line 4: $= \sin 6x$ (double angle).

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4

Hands down. Thank you.

A B C D

EXPLAIN

Error hunt. Compute $\dfrac{d}{dx}\sin^2(3x)$. Which line is the *first* wrong one?

Line 1: $\sin^2(3x) = (\sin 3x)^2$: outside the square, inside $\sin 3x$; so $2\sin 3x \cdot (\sin 3x)'$.
Line 2: $(\sin 3x)' = \cos 3x$.
Line 3: So the derivative is $2\sin 3x \cos 3x$.
Line 4: $= \sin 6x$ (double angle).

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

EXPLAIN

Error hunt. Compute $\dfrac{d}{dx}\sin^2(3x)$. Which line is the *first* wrong one?

Line 1: $\sin^2(3x) = (\sin 3x)^2$: outside the square, inside $\sin 3x$; so $2\sin 3x \cdot (\sin 3x)'$.

Line 2: $(\sin 3x)' = \cos 3x$.

Line 3: So the derivative is $2\sin 3x \cos 3x$.

Line 4: $= \sin 6x$ (double angle).

A. Line 1 **B.** Line 2 **C.** Line 3 **D.** Line 4

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

B. $\sin 3x$ is itself a chain: outside sin, inside $3x$, so $(\sin 3x)' = 3\cos 3x$ — the missing factor of the Lecture 10 warm-up. Line 1 is right, and lines 3 and 4 follow correctly from the wrong line 2: correct algebra carries a wrong line all the way to the end. The true derivative is $2\sin 3x \cdot 3\cos 3x = 3\sin 6x$.

Question 29

The point $(3, 2)$ lies on the curve $x^2 - y^2 = 5$. The slope of the tangent there is

A. $\dfrac{2}{3}$

B. $-\dfrac{3}{2}$

C. 3

D. $\dfrac{3}{2}$

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

The point $(3, 2)$ lies on the curve $x^2 - y^2 = 5$. The slope of the tangent there is

A. $\frac{2}{3}$

B. $-\frac{3}{2}$

C. 3

D. $\frac{3}{2}$

60 sec

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

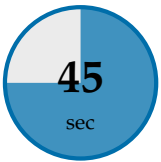
The point $(3, 2)$ lies on the curve $x^2 - y^2 = 5$. The slope of the tangent there is

A. $\frac{2}{3}$

B. $-\frac{3}{2}$

C. 3

D. $\frac{3}{2}$



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

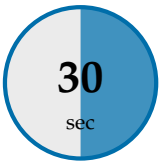
The point $(3, 2)$ lies on the curve $x^2 - y^2 = 5$. The slope of the tangent there is

A. $\frac{2}{3}$

B. $-\frac{3}{2}$

C. 3

D. $\frac{3}{2}$



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

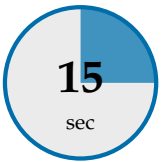
The point $(3, 2)$ lies on the curve $x^2 - y^2 = 5$. The slope of the tangent there is

A. $\frac{2}{3}$

B. $-\frac{3}{2}$

C. 3

D. $\frac{3}{2}$



In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

The point $(3, 2)$ lies on the curve $x^2 - y^2 = 5$. The slope of the tangent there is

A. $\frac{2}{3}$

B. $-\frac{3}{2}$

C. 3

D. $\frac{3}{2}$

0
sec

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

VOTE

The point $(3, 2)$ lies on the curve $x^2 - y^2 = 5$. The slope of the tangent there is

A. $\frac{2}{3}$

B. $-\frac{3}{2}$

C. 3

D. $\frac{3}{2}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

The point $(3, 2)$ lies on the curve $x^2 - y^2 = 5$. The slope of the tangent there is

A. $\frac{2}{3}$

B. $-\frac{3}{2}$

C. 3

D. $\frac{3}{2}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

The point $(3, 2)$ lies on the curve $x^2 - y^2 = 5$. The slope of the tangent there is

A. $\frac{2}{3}$

B. $-\frac{3}{2}$

C. 3

D. $\frac{3}{2}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

The point $(3, 2)$ lies on the curve $x^2 - y^2 = 5$. The slope of the tangent there is

A. $\frac{2}{3}$

B. $-\frac{3}{2}$

C. 3

D. $\frac{3}{2}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B **C** D

VOTE

The point $(3, 2)$ lies on the curve $x^2 - y^2 = 5$. The slope of the tangent there is

A. $\frac{2}{3}$

B. $-\frac{3}{2}$

C. 3

D. $\frac{3}{2}$

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C **D**

VOTE

The point $(3, 2)$ lies on the curve $x^2 - y^2 = 5$. The slope of the tangent there is

A. $\frac{2}{3}$

B. $-\frac{3}{2}$

C. 3

D. $\frac{3}{2}$

Hands down. Thank you.

A B C D

EXPLAIN

The point $(3, 2)$ lies on the curve $x^2 - y^2 = 5$. The slope of the tangent there is

A. $\frac{2}{3}$

B. $-\frac{3}{2}$

C. 3

D. $\frac{3}{2}$

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

EXPLAIN

The point $(3, 2)$ lies on the curve $x^2 - y^2 = 5$. The slope of the tangent there is

A. $\frac{2}{3}$

B. $-\frac{3}{2}$

C. 3

D. $\frac{3}{2}$

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

D. Differentiate as it stands: $2x - 2y\,y' = 0$, so $y' = \frac{x}{y} = \frac{3}{2}$ at $(3, 2)$. B is the circle's $-x/y$ from Lecture 10, copied without differentiating — this curve has a minus sign in it. C is $2x - 2y' = 0$: the y dropped from $\frac{d}{dx}(y^2) = 2y\,y'$, the outside derivative left without its inside. A is the fraction upside down. (Check: $9 - 4 = 5$, so the point is on the curve — always check.)

By heart (1) — the definition and the rules

The definition, both spellings

$f'(a) = \lim_{h \to 0} \dfrac{f(a+h) - f(a)}{h} = \lim_{x \to a} \dfrac{f(x) - f(a)}{x - a}$ — the tangent slope, the instantaneous rate, the velocity: one limit, three names.

By heart (1) — the definition and the rules

The definition, both spellings

$f'(a) = \lim_{h\to 0} \dfrac{f(a+h)-f(a)}{h} = \lim_{x\to a} \dfrac{f(x)-f(a)}{x-a}$ — the tangent slope, the instantaneous rate, the velocity: one limit, three names.

The rules

$(c)' = 0$, $(x^n)' = nx^{n-1}$, $(cf)' = cf'$, $(f \pm g)' = f' \pm g'$, $(fg)' = fg' + gf'$;

$\left(\dfrac{f}{g}\right)' = \dfrac{gf' - fg'}{g^2}$; $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\tan x)' = \sec^2 x$ (sec, csc, cot by the quotient rule);

$\big[f(g(x))\big]' = f'(g(x)) \cdot g'(x)$, $\dfrac{dy}{dx} = \dfrac{dy}{du}\dfrac{du}{dx}$.

By heart (2) — implicit differentiation, the theorems, the habits

Implicit differentiation, two steps

Differentiate both sides with respect to x — every y you differentiate brings a y' — then solve for y'. The answer may contain x and y.

By heart (2) — implicit differentiation, the theorems, the habits

Implicit differentiation, two steps

Differentiate both sides with respect to x — every y you differentiate brings a y' — then solve for y'. The answer may contain x and y.

The theorems you must be able to state, with their hypotheses

Limit laws: if $\lim f$ and $\lim g$ exist, so do the limits of sum, product and (non-zero limit below) quotient; the Squeeze Theorem. **IVT**: f continuous on $[a, b]$, $f(a) \neq f(b)$, N between them $\Rightarrow$ $f(c) = N$ for some c in (a, b). **Differentiable** $\Rightarrow$ **continuous**, with its two-line proof, and $|x|$ against the converse.

By heart (2) — implicit differentiation, the theorems, the habits

Implicit differentiation, two steps

Differentiate both sides with respect to x — every y you differentiate brings a y' — then solve for y'. The answer may contain x and y.

The theorems you must be able to state, with their hypotheses

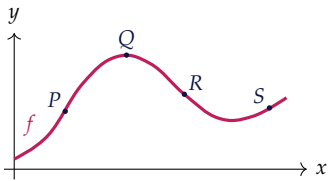
Limit laws: if $\lim f$ and $\lim g$ exist, so do the limits of sum, product and (non-zero limit below) quotient; the Squeeze Theorem. **IVT**: f continuous on $[a, b]$, $f(a) \neq f(b)$, N between them $\Rightarrow$ $f(c) = N$ for some c in (a, b). **Differentiable** $\Rightarrow$ **continuous**, with its two-line proof, and $|x|$ against the converse.

The habits

Rewrite before you reach for a rule. Say "outside" before you differentiate a composition. Check that the point is on the curve. A rate has units. Radians, always.

Question 30

The graph of f is shown, with four points marked. At which point is $f'(x)$ *largest*?



A. P **B.** Q

C. R **D.** S

In your team

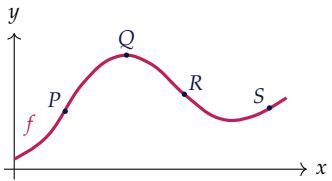
Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

The graph of f is shown, with four points marked. At which point is $f'(x)$ *largest*?

A. P **B.** Q

C. R **D.** S



60 sec

In your team

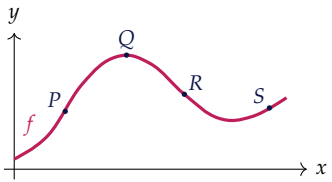
Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

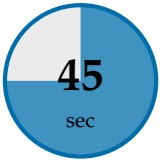
THINK & TALK · 1 minute

The graph of f is shown, with four points marked. At which point is $f'(x)$ *largest*?

A. P **B.** Q

C. R **D.** S





In your team

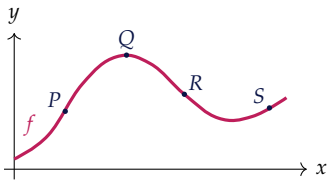
Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

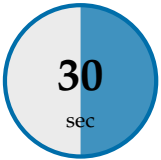
THINK & TALK · 1 minute

The graph of f is shown, with four points marked. At which point is $f'(x)$ *largest*?

A. P **B.** Q

C. R **D.** S





In your team

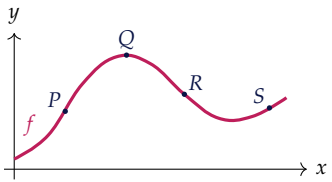
Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

The graph of f is shown, with four points marked. At which point is $f'(x)$ *largest*?

A. P **B.** Q

C. R **D.** S





In your team

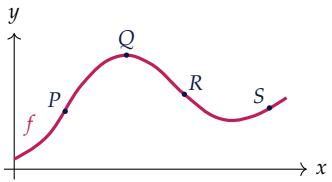
Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

THINK & TALK · 1 minute

The graph of f is shown, with four points marked. At which point is $f'(x)$ *largest*?

A. P **B.** Q

C. R **D.** S



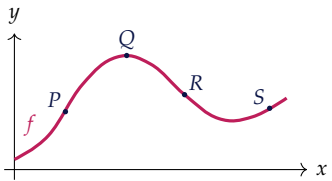
0
sec

In your team

Write your own letter first — then talk. Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

VOTE

The graph of f is shown, with four points marked. At which point is $f'(x)$ *largest*?



A. P **B.** Q

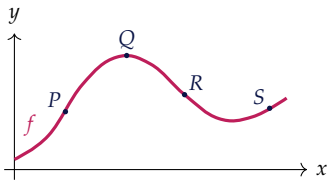
C. R **D.** S

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

The graph of f is shown, with four points marked. At which point is $f'(x)$ *largest*?



A. P **B.** Q

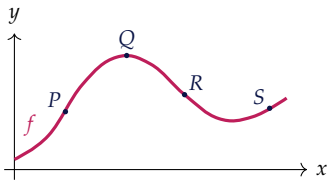
C. R **D.** S

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

The graph of f is shown, with four points marked. At which point is $f'(x)$ *largest*?



A. P **B.** Q

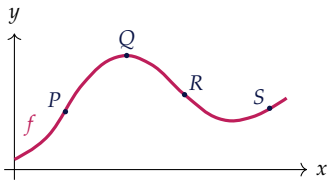
C. R **D.** S

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE

The graph of f is shown, with four points marked. At which point is $f'(x)$ *largest*?



A. P **B.** Q

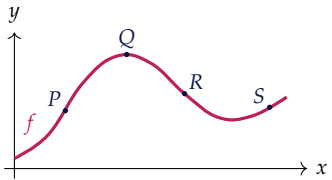
C. R **D.** S

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B **C** D

VOTE

The graph of f is shown, with four points marked. At which point is $f'(x)$ *largest*?



A. P **B.** Q

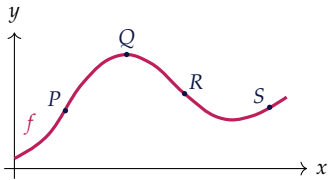
C. R **D.** S

Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C **D**

VOTE

The graph of f is shown, with four points marked. At which point is $f'(x)$ *largest*?



A. P **B.** Q

C. R **D.** S

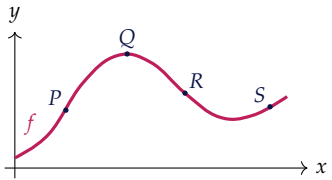
Hands down. Thank you.

A B C D

Question 30

EXPLAIN

The graph of f is shown, with four points marked.
At which point is $f'(x)$ *largest*?



A. P **B.** Q

C. R **D.** S

Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

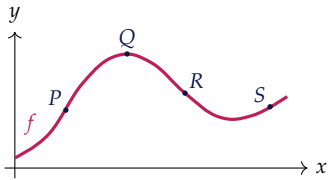
Question 30

EXPLAIN

The graph of f is shown, with four points marked.
At which point is $f'(x)$ *largest*?

A. P **B.** Q

C. R **D.** S



Who explains?

Hands up — one of you gives the reason; I repeat it aloud for everybody.

Answer

A. f' is the *slope*, not the height. At P the graph is climbing steeply; at Q, the highest point, the tangent is horizontal and $f'(Q) = 0$; at R the graph is falling, $f' < 0$; at S it climbs again, but gently. B is Lecture 8's Question 17 in a new dress: where f is largest is not where f' is largest.

Before the tutorial — three things

1. Stewart's Concept Check and True–False Quiz at the end of Chapters 1 and 2 (the items on §1.5–1.8, §2.1–2.6; skip §2.7–2.9) and Chapter 3's items on §3.4 — without notes, then check. Several "false" ones are cousins of today's standing votes.

Before the tutorial — three things

1. Stewart's Concept Check and True–False Quiz at the end of Chapters 1 and 2 (the items on §1.5–1.8, §2.1–2.6; skip §2.7–2.9) and Chapter 3's items on §3.4 — without notes, then check. Several "false" ones are cousins of today's standing votes.
2. Redo the unfinished homework of Tutorials 2–4 — a limit by factoring, a continuity check, a derivative from the definition — with the four steps for each derivative and the three conditions for each continuity check.

Before the tutorial — three things

1. Stewart's Concept Check and True–False Quiz at the end of Chapters 1 and 2 (the items on §1.5–1.8, §2.1–2.6; skip §2.7–2.9) and Chapter 3's items on §3.4 — without notes, then check. Several "false" ones are cousins of today's standing votes.
2. Redo the unfinished homework of Tutorials 2–4 — a limit by factoring, a continuity check, a derivative from the definition — with the four steps for each derivative and the three conditions for each continuity check.
3. One page in your own words: the map of frame 3, with one example of your own in each box. Bring it to the tutorial.

Before the tutorial — three things

1. Stewart's Concept Check and True–False Quiz at the end of Chapters 1 and 2 (the items on §1.5–1.8, §2.1–2.6; skip §2.7–2.9) and Chapter 3's items on §3.4 — without notes, then check. Several "false" ones are cousins of today's standing votes.
2. Redo the unfinished homework of Tutorials 2–4 — a limit by factoring, a continuity check, a derivative from the definition — with the four steps for each derivative and the three conditions for each continuity check.
3. One page in your own words: the map of frame 3, with one example of your own in each box. Bring it to the tutorial.

Next:

Lecture 11 — inverse functions and their derivatives (Stewart §6.1), where $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$ and the chain rule is the reason.

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name; names of those present today.
2. One sentence: *each of you, one line with your initials: "the one kind of question from Lectures 1–10 that I still cannot do is . . . " (a kind of question, not a chapter).*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you.

Give it to the professor or the course assistant as you go. Thank you.